

Introduction to Programming

Lecture 5 — Computers' memory

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Université de Saint-Étienne | Year 2026/2027

-1 Recap' time

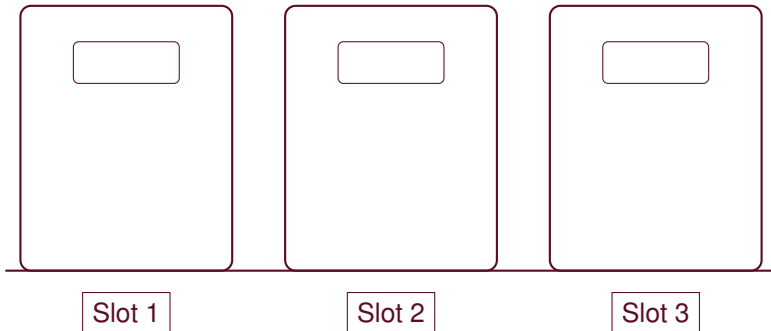
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Assignment review

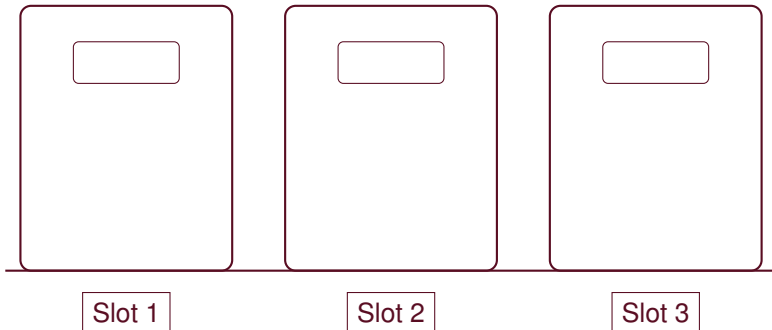
01

**Memory spots, addresses
and pointers**

Boxes in your memory

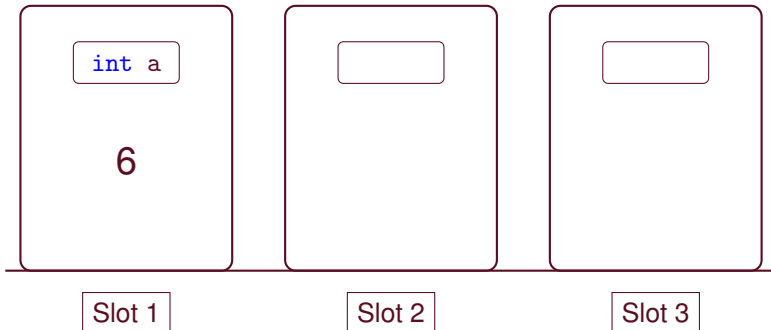


Boxes in your memory



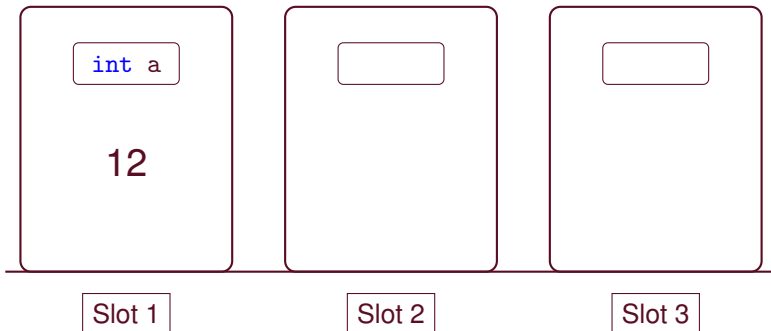
```
int a = 6;
```

Boxes in your memory



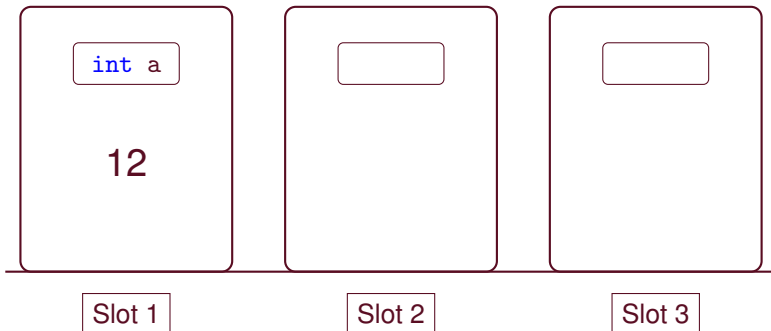
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Boxes in your memory



```
int a = 6;  
a = 12;
```

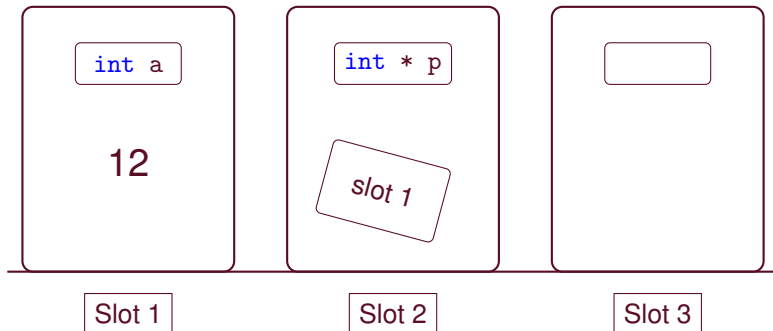
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To declare a pointer, you write the type of the variable you'll store the address of, and add a *.

To get the address of a variable, you write a & before the variable's name.

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To get the address of a variable, you write a & before the variable's name.

```
int a = 12;  
int * p = &a; // p contains the address of a  
// we say that p points to a
```

Reaching the value pointed by a pointer

You can **dereference** a pointer to read the value contained in the variable it is pointing at.

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int a = 12;  
int * p = &a; // p contains the address of a  
std::cout << "a = " << *p;
```

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When initializing a pointer, if you don't assign it a value yourself, it will contain a random memory address. This means the pointer points at a memory spot on your computer that may contain anything. It could contain sensitive data.

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Address zero or `nullptr`

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You set a pointer to point to 0 with either one of the following:

```
int * p = 0;
```

```
int * p = NULL;
```

```
int * p = nullptr;
```

Copying a pointer 1/2

```
int a = 12;  
int b = a;  
a = 4;  
// b still contains 12 because a and b are different boxes
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```
int a = 12, b = 13;
int * p1 = &a;
int * p2 = p1;
*p1 = b;
std::cout << *p2 << std::endl;
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Let's look here

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Practise time!

Data size

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In the same way, a variable uses up program storage / memory. A variable also has a size, that depends on its *type*. This means the boxes on the shelf have *different sizes*.

You can get the size of a variable, that is the memory it uses in a program, with `sizeof`. It will give you the size in bytes.

Incrementing a pointer 1/2

A pointer points to a spot in the program's memory. To increment a pointer is to make it point to the next spot in the program's memory.

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int a;
```



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Incrementing a pointer 1/2

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```
int a;
```



p+1;

p+2;

...

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Incrementing a pointer 2/2

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Look at this.

Why would we want to increment a pointer?

Remember argv?

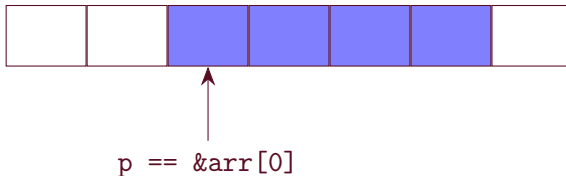
Pointers and arrays

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```
std::array<int, 4> arr {1,2,3,4};  
int * p = arr.data();  
size_t count = arr.size(); // == 4
```



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```
std::array<int, 4> arr
    {1,2,3,4};
int * p = arr.data();
for(int i = 0; i < arr.size();
    ++i){
    std::cout << arr[i]
        << std::endl;
}
```

```
std::array<int, 4> arr
    {1,2,3,4};
int * p = arr.data();
for(int i = 0; i < arr.size();
    ++i){
    std::cout << *(p+i)
        << std::endl;
}
```

C-style arrays

```
<type> arr [] = { ... };
```

C-style arrays are static arrays, very much like `std::array`. The difference is they don't have member functions like `size()`, so you have to keep track of their element count in a separate variable.

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C-style arrays and pointers are most often interchangeable once instantiated.

```
void foo(int * ptr);  
int arr [] = {1,2,3,4};  
foo(arr); // OK
```

Pointer to const

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The word `const` sits *beside the type*.

```
int a = 10;
```

```
const int * p = &a;
```

```
*p = 8; // illegal, p is pointer to const
```

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```
int a = 10, b = 78;
```

```
int * const p = &a;
```

```
p = &b; // illegal, p is a const pointer
```

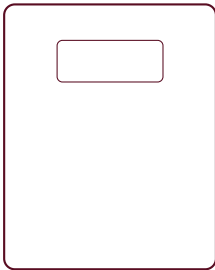
References vs pointers 1/3

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int a = 6;
```

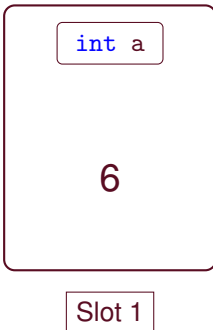


Slot 1

References vs pointers 1/3

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int a = 6;
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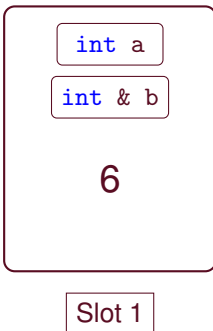


References vs pointers 1/3

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```
int a = 6;
```

```
int & b = a;
```



References vs pointers 2/3

References	Pointers
Have the same address as the variable they denote	Have their own address

References vs pointers 2/3

References	Pointers
Have the same address as the variable they denote	Have their own address
Cannot be empty, thus cannot exist beyond the scope of the variable they denote.	Will contain an address no matter what. If the object they point to is destroyed before them, the pointer becomes invalid but can still exist and be used.

References vs pointers 3/3

Because pointers can become invalid and references can't, the latter is safer to use than the former.

Rule of thumb: whenever possible, if you have the choice between the two, prefer references.

02 Accessing text files

Reading from text files: an example

Let's have a look at an example.

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`std::ifstream` is an **input file stream**. You can use it to **read** from a file.

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`std::ifstream` is an **input file stream**. You can use it to **read** from a file.

You can either read a file line by line with `std::getline`.

Or you can use the operator `>>` to read data into variables. You have to be careful about type (mis-)match.

Writing to text files: an example

Let's have a look at an example.

Writing to text files

`std::ofstream` is an **output file stream**. You can use it to **read** from a file.

Writing to text files

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You can write anything to a file using `<<` just like you'd write to the terminal output with `std::cout << ....`

Practise time!

03

**Reading and modifying
audio files**

What are sound files? Briefly

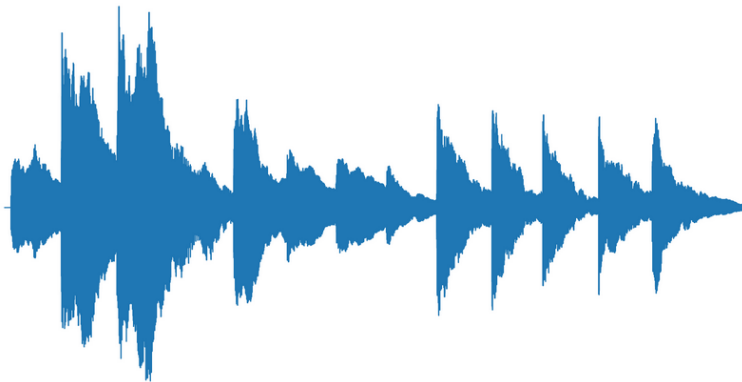


Figure: Audio waveform

Audio files and how to manipulate them

Sound can be represented as a collection of `float` values all in the range $[-1, 1]$.

Audio files and how to manipulate them

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We can use the `libsnfile` library to open, read from and write to audio files.

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See you next time!